MIL-C-60142A(MU)
10 May 1966
SUPERSEDING
MIL-c-60142 (MU)
7 January 1965

MILITARY SPECIFICATION

CONTROL, GIMBAL POSITION: 10527253

1. SCOPE

1.1 This specification covers one type of gimbal positioning control that applies an angular rate to the servo loop of a gyro-stabilized sight assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-T-152	Treatment, Moisture and Fungus-Resistant, of Communications, Electronic and Associated Electrical Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

U.S. Army, Frankford Arsenal

c10527253	Control, Gimbal Position: 10527253

FSC 1270

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PACKAGING DATA SHEET

MIL-P-14232/P10527253 Packaging of Control, Gimbal Position: 10527253

(Copies **of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)**

3. REQUIREMENTS

3.1 First article testing.- Requirements for the submission of first article samples by the contractor for testing shall be as specified in the contract.

3.2 Design and construction.- The gimbal position control shall be designed and constructed in accordance with Drawing C10527253 and the requirements of this specification.

3.2.1 Treatment, moisture and fungus-resistant .- The gimbal position control shall be treated for moisture and fungus resistance in accordance with MIL-T-152.

3.3 General specification.- The contractor shall be responsible for adherence to the following requirements of MIL-F-13926:

(a) Dimensions and tolerances
(b) Inorganic protective surface finishes
(c) Part identification and marking
(d) Workmanship
(e) Soldering electrical connections

3.4 Environmental.

3.4.1 Storage temPerature.- The gimbal position control shall meet the performance requirements of 3.5 at 60° to 90° Fahrenheit (F) after being subjected to ambient temperatures at -80° and at +160°F.

3.4.2 Operating temperature.- The gimbal position control shall meet the performance requirements of 3.5 while exposed to temperatures at -40° and at +150° Fahrenheit (F).

3.4.3 Altitude.- The gimbal position control shall meet the performance requirements of 3.5 at an altitude of 30,000 feet above sea level.

3.4.4 Shock.- The gimbal position control shall withstand impact shocks Up to **15 G's, each shock impulse having a time duration of 11 ± 1 milliseconds. Subsequently, the gimbal position control shall meet the performance requirements of 3.5.**

3.4.5 Vibration.- The gimbal position control shall meet the performance requirements **of 3,5 after being subjected to vibrations up to 5 G's over a frequency range of 5 to 500 cycles per second.**

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3.4.6 Salt-laden air.- The gimbal position control shall meet the performance requirements of 3.5 after being subjected to the following salt-laden conditions.

(a) Salt solution concentration: 5%
(b) Specific gravity: 1.0268 to 1.0413
(c) pH from 6.5 to 7.2 at temperatures from 93° to 97°F

3.5 Performance.

3.5.1 Output Per axis.- When excited by an input voltage of 26 volts alternating current (Vac)~ 1% at 400 cps each of the two output axes shall have a sensitivity of 0.59 volts per pound root mean square (Vrms) ± 20% at 400 cps at Oo and 180° phase for a O to 10 pound force directed along the axis in both directions. The sensitivity shall be the slope of the most favorable straight line drawn for the linearity test.

3.5.2 Linearity.- The voltage versus applied force characteristic shall be within 5.0% of the most favorable straight line drawn through the data for each axis and for both phases.

3.5.3 Hysteresis.- The hysteresis at the null for 10 pound force in opposite directions on the lever shall be less than 1%.

3.6 Radio interference suppression.- Interference control requirements shall conform to MIL-I-6181.

4, QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article testing.- When the contract specifies testing of preproduction or initial production samples, the following paragraphs shall apply as applicable.

4.2.1 Preproduction test.- Preproduction test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, two preproduction samples shall be submitted to the procuring agency for approval of design and construction features. Preproduction samples shall meet all inspection provisions of 4.3, 4.4 and 4.6 as well as any additional tests that may be specified in the contract.

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4.2.2 Initial production test.- Initial production test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, samples shall be the first two production units and shall have been manufactured using the same materials, processes and procedures as will be used in regular production. Initial production testing shall be in accordance with 4.3, 4.4 and 4.6 and shall be witnessed by the Government representative for all examinations and tests. The Government reserves the right to conduct any or all examinations or tests as **may be** deemed necessary.

4.2.3 First article disapProval.- Failure of any one of a first article sample submitted shall be cause for disapproval by the Government and subsequent action as required by the contract clauses applicable to first article testing.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the Contracting Officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents (IEL), when provided with the technical data package (TDP). In the absence of SQAP'S the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 First article tests.- Unless otherwise specified, the requirements and tests of Table I shall be conducted on first article tests only. In the event that the processes, materials or other changes in design or manufacture occur subsequent to first article approval, the first items so produced shall be submitted to the Government in accordance with 4.2 for approval prior to resumption of production.

TABLE I

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	**Altitude**	**3.4.3**	**4.6.2.3**
302.	**Shock**	**3.4.4**	**4.6.2.4**
303.	**Vibration**	**3.4.5.**	**4.6.2.5**
304.	**Salt laden air**	**3.4.6**	**4.6.2.6**
305.	**Radio interference suppression**	**3.6**	**4.6.3.3**

4.3.2.3 **Final acceptance inspection.- Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables 11 and III herein. Examination and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The** tabulated **classification of defects in Tables 11 and III shall constitute the minimum**

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inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE 11 CLASSIFICATION OF DEFECTS

CLASS	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR : AQL 1.5 Percent Defective		
101. Output per axis	**3.5.1**	**4.6.3.1**
102. Linearity	**3.5.2** .	**4.6.3.1**
103. Hysteresis	**3.5.3**	**4.6.3.2**
104. Moisture and fungus resistance	**3.2.1**	**4.6.1**
MINOR: NONE		

NOTE : The tests in Table II shall be conducted at a temperature between 60° and 90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special samPling.- Three controls shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table III.

TABLE 111

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
401. Design and construction	3.2	Applicable drawing - visual
402. General specification	3.3	MIL-F-13926 - visual
4030 Temperature (storage)	3.4.1	4,6.2.1
404. Temperature (operating)	3.4.2	4.6.2.2

4.4.1 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equiPment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

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4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents when provided with the technical data package (TDP),

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive. rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test methods and Procedures.

4.6.1 Treatment, moisture and fungus-resistant. - The gimbal position control shall be tested in accordance with MIL-T-152 to insure compliance with 3.2.1.

4.6.2 Environmental.

4.6.2.1 Storage temPerature.- The gimbal position control, while in a non-operating condition, shall be subjected to the Temperature Cycling Test, Procedure II of MIL-F-13926 using Conditions A and H. Upon stabilization at ambient room temperature, the gimbal position control shall meet the performance requirements specified in 3.5.

4.6.2.2 Operating temPerature.- The gimbal position control shall be subjected to the Temperature Cycling Test, Procedure I of MIL-F-13926 using Conditions C and G. During temperature stabilization at Conditions C, G and room ambient, the gimbal position control shall meet the performance requirements specified in 3.5.

4.6.2.3 **Altitude test.- The gimbal position control shall be placed within a test chamber and the absolute internal pressure reduced** to **8.88 inches of mercury (corresponding to an altitude of 30,000 feet above sea level). After sufficient time has elapsed to allow all entrapped air to escape, the gimbal position control shall meet the requirements specified in 3.5.**

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4.6.2.4 Shock test.- The gimbal position control shall be tested in accordance with MIL-F-13926, Procedure II. The gimbal position control shall not suffer damage or subsequently fail to provide the performance specified in 3.5.

4.6.2.5 Vibration test.- With the gimbal position control mounted and operating at room ambient conditions, the table shall be vibrated along a horizontal direction with the frequency varying between 5 and 500 Cps. For frequencies between 5 and 20 cps, the double amplitude shall be 0.1 inch. For frequencies between 20 and 33 cps, the amplitude shall be varied so as to maintain a 2 G acceleration. For frequencies between 33 and 52 cps, the double amplitude shall be 0.036 inch. For frequencies between 52 and 500 cps, the amplitude shall be varied so as to maintain a 5 G acceleration. The rate of frequency change shall be logarithmic and shall be such that a complete cycle (5-500-5) will consume approximately 15 minutes. The test shall continue for four 15-minute cycles for a total of 1 hour. (When provisions for logarithmic cycling is not available, a linear rate of frequency change may be used.) The cycling test shall be repeated with the axis of vibration changed 90 degrees in the horizontal direction and then with the axis of vibration changed to vertical. At the completion of each direction of vibration, the gimbal position control shall be closely inspected for any evidence of mechanical failures loose or broken parts. At end of test, the gimbal position control shall meet the performance requirements of 3.5.

4.6.2.6 Salt laden air test.- The gimbal position control shall be tested in accordance with Method 101B of MIL-STD-202, Condition A. After exposure, the gimbal position control shall not suffer any damage or subsequently fail to provide the performance specified in 3.5.

4.6.3 Performance.

4.6.3.1 Linearity and maximum output test.- Install the gimbal position control in a testing device consisting of two O to 10 lb scales, adjustable springs and a protractor-type angular chart for establishing orthogonal axes as showm in Figure 1. Set the index mark (scribed on the gimbal position control housing) to the zero of the angular scale. Apply 26 Vac rms ± 1% at 400 cps excitation to the unit. Next, **manually adjust** the tension on one spring for scale readings of 2, 4, 6, 8 and 10 Ibs. For each force input, read the output voltage (Figure 2). Repeat these readings for pound forces measured on the other scale. During these tests, verify, by means of a phasemeter or oscilloscope, that 0° and 180 phase relations about the neutral exist between each output voltage and the excitation voltage of the gimbal position control. Rotate the gimbal position control in the holding device so that the scribed line **is** coincident with the 90° line of the orientation indicator. Repeat input-output function for both polarity of deflections of the gimbal position control. Again, during these tests, verify by means of a phasemeter or oscilloscope, that 0° and 180° phase relations about the neutral exist between each output voltage and the excitation voltage of the gimbal position control. Plot the input-output characteristics to verify compliance with 3.5.1 and 3.5.2.

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4.6.3.2 Hysteresis test.- For each of the orthogonal axes, apply first a 10 lb. force in one direction then in the other. Release the handle after each application of the force and measure the output at neutral to verify compliance with 3.5.3.

4.6.3.3 Radio interference test.- The gimbal position control shall be tested in accordance to MIL-I-6181 to insure conformance with 3.6.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P10527253. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The Gimbal Position Control: 10527253 is part of Antioscillation Sighting System: XM58 employed in a UH-lB Helicopter. The fixed angular rate applied to the servo loop by the gimbal position control is intended to be used while the missile is in flight. This function allows the helicopter some translational motion freedom. Under this type of vehicle motion, the target or missile can begin to drift out of range of the optics. The operator can bring the target back into the field of view by manipulation of the gimbal position control and the missile by the missile controls.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of preservation, packaging, and packing (see 5.1).
(c) Applicable packaging data sheet number (see 5.1).
(d) Applicable stock number.

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Project No. 1270-A032Y

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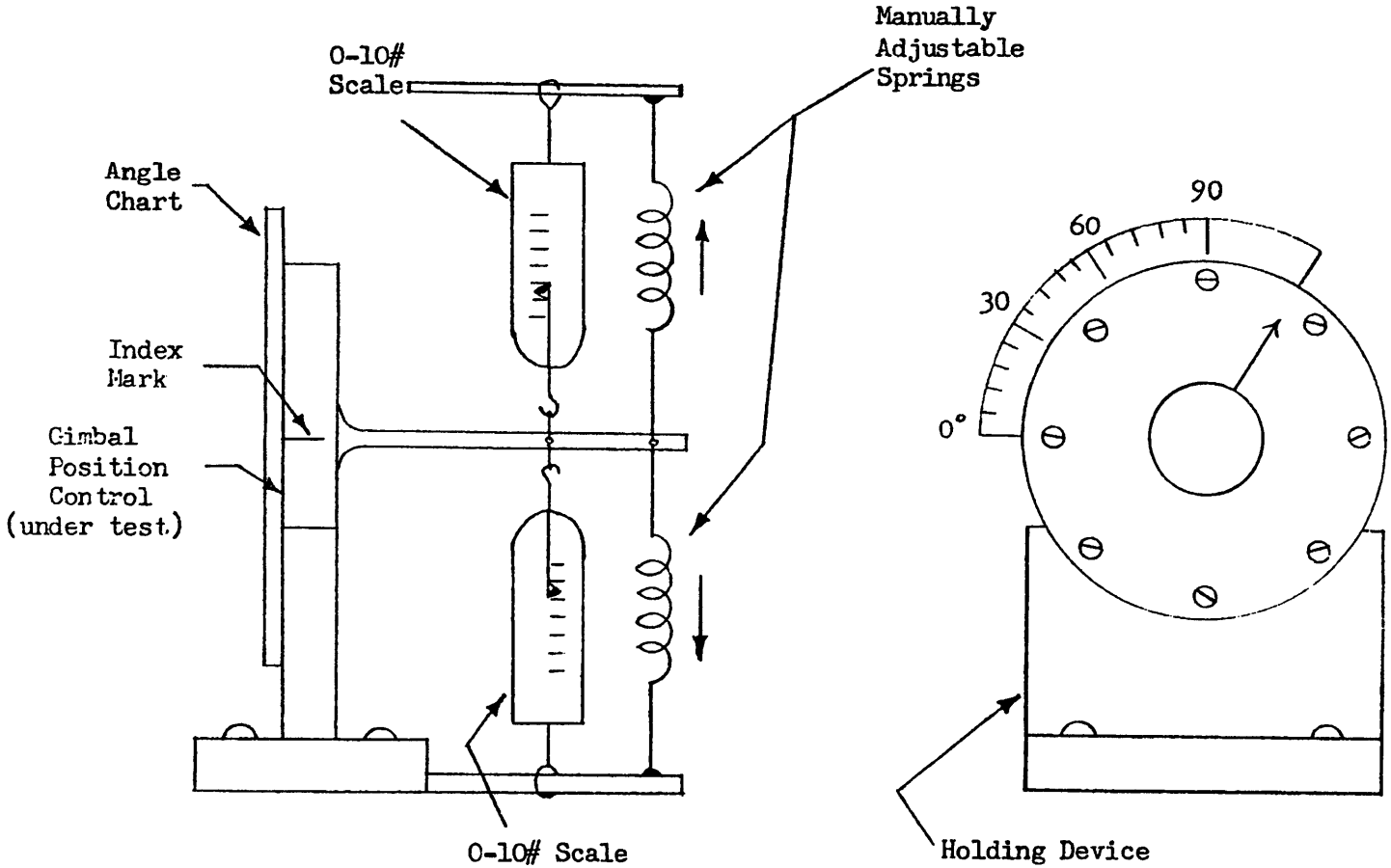


FIGURE 1. Test Setup for Gimbal Position Control

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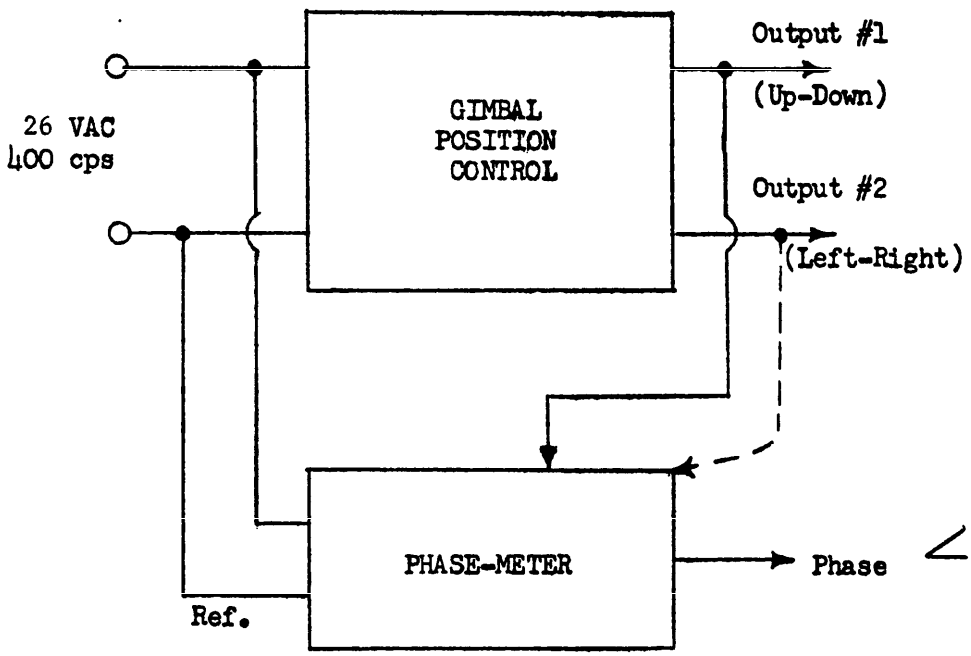


FIGURE 2. Test Schematic for Phase Check

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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